

Collections

Create an Employee class with following properties

EmployeeName as string

EmployeeID as int

Salary as double

Create another class called EmployeeDAL which has the following methods.

1. Create an ArrayList within this class

2. Define the below methods

a. public bool AddEmployee(Employee e)

b. public bool DeleteEmployee(int id)

c. public string SearchEmployee(int id) : Returns the employee name if found
else return null

d. public Employee[] GetAllEmployeesistAll()

Use an ArrayList to store the employees

Write a test program to test all the functionalities.

```
using System;
```

```
using System.Collections;
```

```
class Employee
```

```
{
```

```
    public string EmployeeName;
```

```
    public int EmployeeID;
```

```
    public double Salary;
```

```
    public Employee(string employeeName, int employeeID, double salary)
```

```
    {
```

```
        EmployeeName = employeeName;
```

```

        EmployeeID = employeeID;
        Salary = salary;
    }
}
class EmployeeDAL
{
    ArrayList employees = new ArrayList();

    public bool AddEmployee(Employee e)
    {
        employees.Add(e);
        return true;
    }

    public bool DeleteEmployee(int id)
    {
        for (int i = 0; i < employees.Count; i++)
        {
            Employee e = (Employee)employees[i];

            if (e.EmployeeID == id)
            {
                employees.RemoveAt(i);
                return true;
            }
        }
        return false;
    }

    public string SearchEmployee(int id)

```

```

{
    foreach (Employee e in employees)
    {
        if (e.EmployeeID == id)
        {
            return e.EmployeeName;
        }
    }
    return null;
}

public Employee[] GetAllEmployeesListAll()
{
    Employee[] allEmployees = new Employee[employees.Count];
    for (int i = 0; i < employees.Count; i++)
    {
        allEmployees[i] = (Employee)employees[i];
    }
    return allEmployees;
}
}

class Program
{
    static void Main()
    {
        EmployeeDAL dal = new EmployeeDAL();
        Employee e1 = new Employee("Pooja", 101, 50000);
        Employee e2 = new Employee("Rahul", 102, 60000);
        Employee e3 = new Employee("Anu", 103, 55000);
    }
}

```

```
dal.AddEmployee(e1);
dal.AddEmployee(e2);
dal.AddEmployee(e3);
Console.WriteLine("Employees added successfully.");
string name = dal.SearchEmployee(102);
if (name != null)
{
    Console.WriteLine("Employee found: " + name);
}
else
{
    Console.WriteLine("Employee not found.");
}
bool deleted = dal.DeleteEmployee(103);
if (deleted)
{
    Console.WriteLine("Employee deleted successfully.");
}
else
{
    Console.WriteLine("Employee not found.");
}
Console.WriteLine();
Console.WriteLine("All Employees:");
Employee[] allEmployees = dal.GetAllEmployeesListAll();
foreach (Employee e in allEmployees)
{
    Console.WriteLine(
```

```
        "ID: " + e.EmployeeID +  
        ", Name: " + e.EmployeeName +  
        ", Salary: " + e.Salary  
    );  
}  
}  
}
```

OUTPUT: Employees added successfully.

Employee found: Rahul

Employee deleted successfully.

All Employees:

ID: 101, Name: Pooja, Salary: 50000

ID: 102, Name: Rahul, Salary: 60000